

OSGi Working Group
OSGi Specification repo for
`org.osgi.service.onem2mRe-`
`lease-1.0.1-SNAPSHOT`

September 2026



Copyright © 2000, 2026 Eclipse Foundation

LICENSE

Eclipse Foundation Specification License – v1.0

By using and/or copying this document, or the Eclipse Foundation document from which this statement is linked, you (the licensee) agree that you have read, understood, and will comply with the following terms and conditions:

Permission to copy, and distribute the contents of this document, or the Eclipse Foundation document from which this statement is linked, in any medium for any purpose and without fee or royalty is hereby granted, provided that you include the following on ALL copies of the document, or portions thereof, that you use:

- link or URL to the original Eclipse Foundation document.
- All existing copyright notices, or if one does not exist, a notice (hypertext is preferred, but a textual representation is permitted) of the form: "Copyright © [date-of-document] Eclipse Foundation, Inc. <<url to this license>>"

Inclusion of the full text of this NOTICE must be provided. We request that authorship attribution be provided in any software, documents, or other items or products that you create pursuant to the implementation of the contents of this document, or any portion thereof.

No right to create modifications or derivatives of Eclipse Foundation documents is granted pursuant to this license, except anyone may prepare and distribute derivative works and portions of this document in software that implements the specification, in supporting materials accompanying such software, and in documentation of such software, PROVIDED that all such works include the notice below. HOWEVER, the publication of derivative works of this document for use as a technical specification is expressly prohibited.

The notice is:

"Copyright © [date-of-document] Eclipse Foundation. This software or document includes material copied from or derived from [title and URI of the Eclipse Foundation specification document]."

Disclaimers

THIS DOCUMENT IS PROVIDED "AS IS," AND THE COPYRIGHT HOLDERS AND THE ECLIPSE FOUNDATION MAKE NO REPRESENTATIONS OR WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, OR TITLE; THAT THE CONTENTS OF THE DOCUMENT ARE SUITABLE FOR ANY PURPOSE; NOR THAT THE IMPLEMENTATION OF SUCH CONTENTS WILL NOT INFRINGE ANY THIRD PARTY PATENTS, COPYRIGHTS, TRADEMARKS OR OTHER RIGHTS.

THE COPYRIGHT HOLDERS AND THE ECLIPSE FOUNDATION WILL NOT BE LIABLE FOR ANY DIRECT, INDIRECT, SPECIAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF ANY USE OF THE DOCUMENT OR THE PERFORMANCE OR IMPLEMENTATION OF THE CONTENTS THEREOF.

The name and trademarks of the copyright holders or the Eclipse Foundation may NOT be used in advertising or publicity pertaining to this document or its contents without specific, written prior permission. Title to copyright in this document will at all times remain with copyright holders.

Preface

Implementation Requirements

An implementation of a Specification: (i) must fully implement the Specification including all its required interfaces and functionality; (ii) must not modify, subset, superset or otherwise extend the OSGi Name Space, or include any public or protected packages, classes, Java interfaces, fields or methods within the OSGi Name Space other than those required and authorized by the Specification. An implementation that does not satisfy limitations (i) (ii) is not considered an implementation of the Specification and must not be described as an implementation of the Specification. "OSGi Name Space" shall mean the public class or interface declarations whose names begin with "org.osgi" or any recognized successors or replacements thereof. An implementation of a Specification must not claim to be a compatible implementation of the Specification unless it passes the Technology Compatibility Kit ("TCK") for the Specification.

Feedback

This specification can be downloaded from the OSGi Documentation web site:

<https://docs.osgi.org/specification/>

Comments about this specification can be raised at:

<https://github.com/osgi/osgi/issues>

Table of Contents

1	Introduction	5
1.1	Reader Level	5
1.2	Version Information	5
1.3	References	5
153	Service Layer API for oneM2M™	7
153.1	Introduction of oneM2M	7
153.2	Application Portability Problem of oneM2M	7
153.3	Introduction of Service Layer API for oneM2M	7
153.4	Essentials	8
153.5	Entities	8
153.6	oneM2M ServiceLayer	9
153.7	NotificationListener	10
153.8	DTO	12
153.9	Security	12
153.10	org.osgi.service.onem2m	13
153.11	org.osgi.service.onem2m.dto	15
153.12	References	35

Licensed under the Eclipse Foundation Specification License – v1.0. Copyright © Contributors to the Eclipse Foundation

DRAFT

1 Introduction

This document contains a specification developed by the OSGi working group.

1.1 Reader Level

This specification is written for the following audiences:

- Application developers
- Framework and system service developers (system developers)
- Architects

This specification assumes that the reader has at least one year of practical experience in writing Java programs. Experience with embedded systems and server-environments is a plus. Application developers must be aware that the OSGi environment is significantly more dynamic than traditional desktop or server environments.

System developers require a very deep understanding of Java. At least three years of Java coding experience in a system environment is recommended. A Framework implementation will use areas of Java that are not normally encountered in traditional applications. Detailed understanding is required of class loaders, garbage collection, Java 2 security, and Java native library loading.

Architects should focus on the introduction of each subject. This introduction contains a general overview of the subject, the requirements that influenced its design, and a short description of its operation as well as the entities that are used. The introductory sections require knowledge of Java concepts like classes and interfaces, but should not require coding experience.

Most of these specifications are equally applicable to application developers and system developers.

1.2 Version Information

Each OSGi specification has an independent version, and packages defined by those specifications have their own versions as well. Specification and package versions follow semantic versioning rules.

1.3 References

- [1] *OSGi Specifications*
<https://docs.osgi.org/specification/>

DRAFT

Licensed under the Eclipse Foundation Specification License – v1.0. Copyright © Contributors to the Eclipse Foundation

153 Service Layer API for oneM2M™

Version 1.0

153.1 Introduction of oneM2M

oneM2M™ is a standard for IoT platform, which is standardized by oneM2M partnership project. oneM2M defines set of functionalities that are commonly used in IoT applications, which is called Common Services Function (CSF). The implementation of the CSF is provided by Communication Service Entity (CSE). oneM2M also defines the interface to use the CSF with REST oriented API that consist of limited types of operation (CREATE, RETRIEVE, UPDATE, DELETE, NOTIFY) on many types of resources. Applications of oneM2M use the interface to communicate with CSEs. In a system managed by a single service provider, multiple CSEs can exist and they form tree structure. The root CSE is called Infrastructure Node CSE (IN-CSE). Each application connects to one of CSEs in the system. CSEs have routing capability and application can send request to any CSEs in the system through the directly-connected CSE.

One of characteristic aspects of oneM2M is to allow multiple protocols and serialization formats for messages. Currently specified protocols are HTTP, CoAP, MQTT and WebSocket, and specified serialization are XML, JSON and CBOR (Concise Binary Object Representation). To make specification coherent, oneM2M specifications are separated into abstract level and concrete level. As abstract level, TS-0001 defines the oneM2M architecture and resource types and TS-0004 defines data procedures and data structures. As concrete level, TS-0008, TS-0009, TS-0010, and TS-0020 define concrete protocol which are mapped to model of the abstract level. Here, the interface defined in abstract level, which independent on concrete protocols, is regarded as oneM2M Service Layer.

oneM2M Partners Type 1 (ARIB, ATIS, CCSA, ETSI, TTA, TSDSI, TTA, TTC) register oneM2M trademarks and logos in their respective jurisdictions.

153.2 Application Portability Problem of oneM2M

One of potential problems is application portability. oneM2M specifies protocol based interfaces, but doesn't specify a programming level API. Without a standardized API, application program tends to be built tightly coupled with the libraries handling the communication method (combination of protocol and serialization) that is initially intend to use. In that case it would be hard to operate the application in another environment where different communication method is used; basically it is required to modify the application drastically. oneM2M could introduce segmentation of ecosystem within oneM2M compliant applications due to the lack of application portability.

153.3 Introduction of Service Layer API for oneM2M

This chapter provides interface to oneM2M applications for communicating communicate CSE at Service Layer of oneM2M. The providing API is protocol and serialization agnostic for preventing the problem above. Once application developer write code, it can be run in other environment where different communication method is used.

Another benefit of the service is reduction of computational resources, typically latency of execution in a certain cases, where both application and CSE is implemented on OSGi framework. In that case, it is possible to reduce execution time for serialization/deserialization of data, context-switch of applications, compared to the case where they communicate with a certain communication protocol.

153.4 Essentials

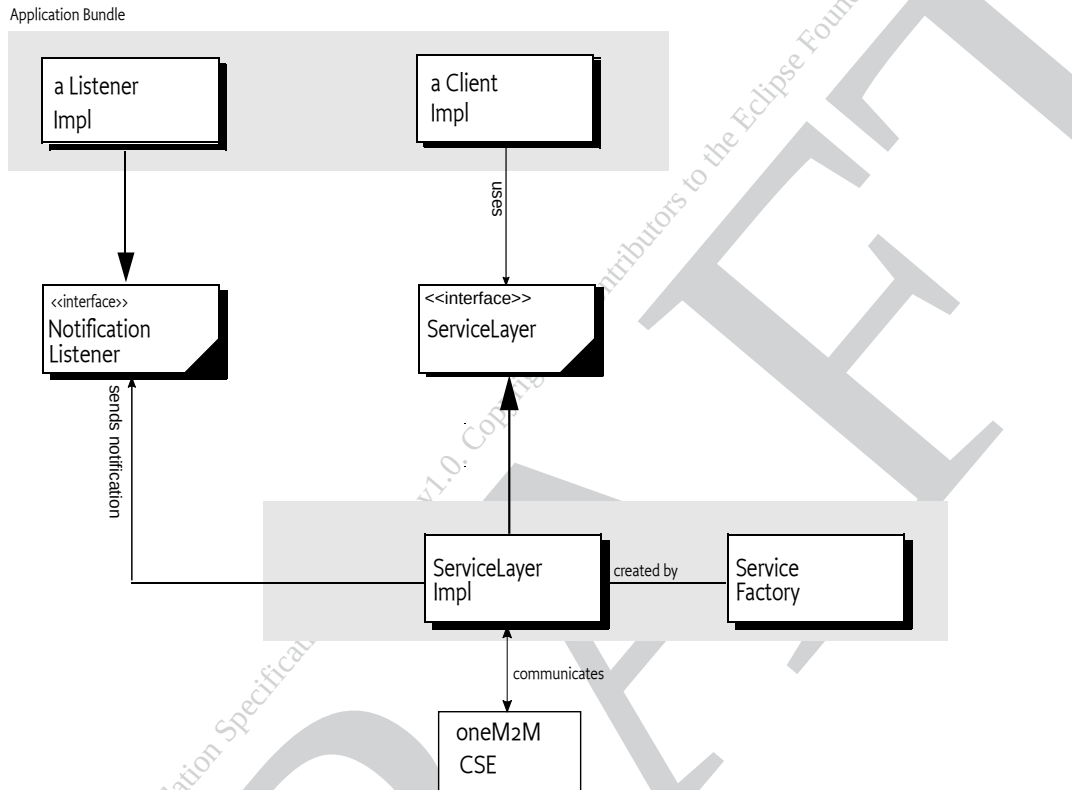
- *Protocol Agnostic* - API is independent on protocol that are used in communications. oneM2M specifies multiple protocols, which are HTTP, CoAP, MQTT and WebSocket. conversion operations.
- *Serialization Agnostic* - API is independent on serialization that are actually used in communications. oneM2M specifies multiple serializations, which are XML, JSON and CBOR .
- *Support of synchronous and asynchronous method call* - API allows both of calling manners.
- *Use of Data Transfer Object (DTO)* - DTO is used as parameters passing on API. Since oneM2M defines many types, concrete DTOs are specified for the higher level structure, and for lower structures generic DTO is used.
- *Low level and high level operations* - API allows for applications to use both low level operation and high level operation, where low level operation allows all possible oneM2M operations and high level operation allows resource level operations, which are create, retrieve, update, and delete.

153.5 Entities

The following entities are used in this specification:

- *Application Bundle* - Application, which use oneM2M CSE's capability. This specification assumes that an application bundle consists an oneM2M application.
- *ServiceLayer* - This is the API used by oneM2M applications.
- *NotificationListener* - Listener Interface, which is necessary to implement by oneM2M applications, when then need to received notifications.
- *ServiceLayer Implementation Bundle* - Bundle providing implementation of ServiceLayer and its ServiceFactory.
- *oneM2M CSE* - oneM2M's Server. It may exist remotely or locally.

Figure 153.1 Entity overview of Service Layer API for oneM2M



153.6 oneM2M ServiceLayer

oneM2M ServiceLayer is the interface used by an application for sending request and get response as return method. It contains low level API and high level API.

request() method allows very raw data type access and it enables all possible message exchanges among oneM2M entities. This is called the low level API. The method takes requestPrimitive as an argument and returns responseRequest. It allows all possible operation of oneM2M. For the return type, OSGi Promise (???) is used for allowing synchronous and asynchronous calling manner.

The low level API may be redundant to application developers, because they need to write composition of requestPrimitive and decomposition of responsePrimitive. Following methods allow application developers to develop application with less lines of code. They provides higher level of abstraction; operation level of resource such as create, retrieve, update, delete, notify and discovery. They cover typical oneM2M operations but do not cover all of possible messages of oneM2M.

Implementation of these high level API automatically inserts 'requestID' and 'from' parameter to RequestDTO.

Following example shows temperature measurement application using container resource and contentInstance resource.

```
ServiceReference<ServiceLayer> sr = bc
    .getServiceReference(ServiceLayer.class);
```

```

ServiceLayer sl = bc.getService(sr);

ResourceDTO container = new ResourceDTO();
container.resourceType = Constants.RT_contentInstance;
container.resourceName = "temperatureStore";
sl.create("/CSE1/csebase", container).getValue();

ScheduledExecutorService service = Executors
    .newSingleThreadScheduledExecutor();

AtomicInteger count = new AtomicInteger(0);

service.scheduleAtFixedRate(() -> {
    ResourceDTO instance = new ResourceDTO();
    instance.resourceType = Constants.RT_contentInstance;
    instance.resourceName = "instance" + count.getAndIncrement();
    instance.attribute = new HashMap<String, Object>();
    instance.attribute.put("content", measureTemperature());

    sl.create("/CSE1/csebase/temperatureStore", instance);
}, 0, 60, TimeUnit.SECONDS);

```

Following example shows visualizing application of temperature data.

```

ServiceReference<ServiceLayer> sr = (ServiceReference<ServiceLayer>) bc
    .getServiceReference("org.osgi.service.onem2m.ServiceLayer");
ServiceLayer sl = bc.getService(sr);

FilterCriteriaDTO fc = new FilterCriteriaDTO();
fc.createdAfter = "20200101T120000";
fc.createdBefore = "20200101T130000";
List<Integer> resourceTypes = new ArrayList<Integer>();
resourceTypes.add(Constants.RT_contentInstance);
fc.resourceType = resourceTypes;
fc.filterOperation = FilterOperation.AND;

List<String> l = sl.discovery("/CSE1/csebase/temperatureStore", fc)
    .getValue();
List<Pair<String, Double>> renderData = new ArrayList<Pair<String, Double>>();
for (String uri : l) {
    ResourceDTO resource = sl.retrieve(uri).getValue();
    renderData.add(new Pair<String, Double>(resource.creationTime,
        (Double) resource.attribute.get("content")));
}

renderService(renderData);

```

153.7 NotificationListener

NotificationListener is an interface for receiving oneM2M notification. An application that needs to receive oneM2M notifications must implement the interface and register it to the OSGi registry.

A ServiceLayer Implementation Bundle must call the notify() method of the NotificationListener, when it receives notification from CSE. In notification, target address is designated by AE-ID. The ServiceLayer Implementation Bundle finds the corresponding instance of the NotificationListener by checking its registerer bundle and checking internal mapping table of AE-ID and application bundle.

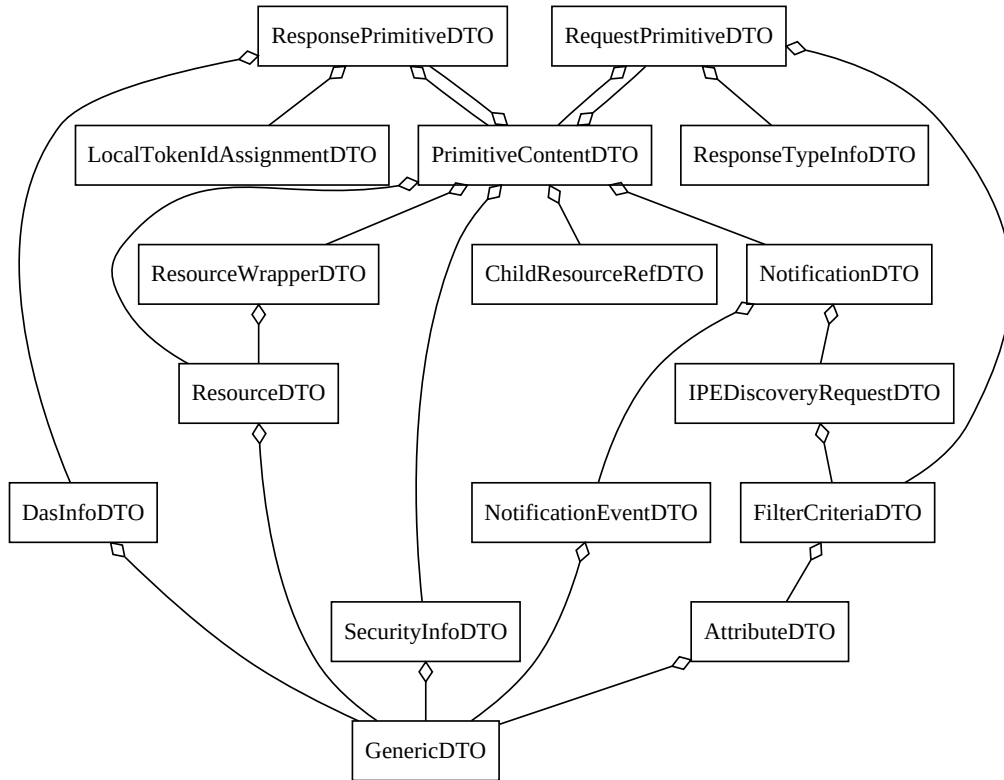
```
public class MyListener implements NotificationListener {
    public void notified(RequestPrimitiveDTO request){
        NotificationDTO notification = request.content.notification;
        NotificationEventDTO event = notification.notificationEvent;
        Object updatedResource = event.representation;
        NotificationEventType type = event.notificationEventType;
        if( type == NotificationEventType.update_of_resource ){
            // check updated resource, execute some actions.
        }
    }
}

@Activate
public void activate(BundleContext bc) {
    NotificationListener l = new MyListener();
    bc.registerService(NotificationListener.class, l, null);
}
}
```

153.8 DTO

OSGi DTOs are used for representing data structured passed on the API. Some of the data structures, which are directly referred from API or in small number of hops, are specified with concrete field names. The following figure shows DTOs with concrete field names, and reference relationship of class. Following DTO's rule, instances must not have loop reference relationship.

Figure 153.2 DTOs representing high level structures



ResourceDTO represents oneM2M resource. ResourceDTO has both fields with concrete names and a field (named as attribute) for having sub-elements in generic manner. All of fields of the ResourceDTO represent attributes. Most of attributes have a primitive type and part of attributes have structured value. For structured value, if it possible defined concrete DTOs must be used, otherwise GenericDTO must be used.

oneM2M specifies two types of key names for representing name of resources, attributes, and elements of data structure, which are long name and short name. Long name is human readable representation, for example "resourceID", meanwhile short name is compact representation for minimizing network transfer, consist with typically 2-4 alphabetical characters, for example "ri". All field names in concrete DTOs are based on long name. Long name should be used for key names of GenericDTO and attribute names of ResourceDTO.

153.9 Security

Implementation of ServiceLayer may use credentials on behalf of application bundles on the communication with oneM2M CSE. So ServiceLayer Implementation should pass the service reference

of ServiceLayer to only the proper application bundle. Use of ServiceFactory is to realize this. Application Bundles should not pass the service reference to other application bundles.

How to configure those credentials is left to developer of ServiceLayer Implementation, and it is out of scope the specification.

153.10 org.osgi.service.onem2m

Service Layer API for oneM2M Specification Package Version 1.0.

Bundles wishing to use this package must list the package in the Import-Package header of the bundle's manifest. This package has two types of users: the consumers that use the API in this package and the providers that implement the API in this package.

Example import for consumers using the API in this package:

```
{@code Import-Package: org.osgi.service.onem2m; version="[1.0,2.0]"}
```

Example import for providers implementing the API in this package:

```
{@code Import-Package: org.osgi.service.onem2m; version="[1.0,1.1]"}
```

153.10.1 Summary

- `NotificationListener` - Interface to receive notification from other oneM2M entities.
- `OneM2MException` - General Exception for oneM2M.
- `ServiceLayer` - Primary Interface for an oneM2M application entity to send request and get response to/from other oneM2M entity.

153.10.2 public abstract NotificationListener

Interface to receive notification from other oneM2M entities.

Application that receives notification must implement this interface and register to OSGi service registry. No service property is required.

153.10.2.1 public abstract void notified(RequestPrimitiveDTO request)

- request* request primitive
- receive notification.

153.10.3 public class OneM2MException extends IOException

General Exception for oneM2M.

153.10.3.1 public OneM2MException(String message, int errorCode)

- message* The exception message.
- errorCode* The exception error code.
- Construct a OneM2MException with a message and an error code.

153.10.3.2 public int getErrorCode()

- Return the error code for the exception.
- Returns* The error code for the exception.

153.10.4 public abstract ServiceLayer

Primary Interface for an oneM2M application entity to send request and get response to/from other oneM2M entity.

It contains low level API and high level API. The only low level method is request() and other methods are categorized as high level API.

Provider Type Consumers of this API must not implement this type

153.10.4.1 public abstract Promise<ResourceDTO> create(String uri, ResourceDTO resource)

uri URI for parent resource of the resource being created.

resource resource data

- create resource

The create() method is a method to create new resource under specified uri. The second argument resource is expression of resource to be generated. The resourceType field of the resourceDTO must be assigned. For other fields depends on resource type. Section 7.4 of TS-00004 specifies the optionalities of the fields.

Returns Promise of created resource

153.10.4.2 public abstract Promise<Boolean> delete(String uri)

uri target URI for deleting resource

- delete resource

delete resource on the URI specified by uri argument.

Returns promise of execution status

153.10.4.3 public abstract Promise<List<String>> discovery(String uri, FilterCriteriaDTO fc)

uri URI for resource tree to start the search

fc filter criteria selecting resources

- find resources with filter condition specified in fc argument.

Discovery Result Type is kept as blank and default value of target CSE is used for the parameter.

Returns list of URIs matching the condition specified in fc

153.10.4.4 public abstract Promise<List<String>> discovery(String uri, FilterCriteriaDTO fc, DesiredIdentifierResultType drt)

uri URI for resource tree to start the search

fc filter criteria

drt Discovery Result Type (structured/unstructured)

- find resources with filter condition specified in fc argument.

With this method application can specify desired identifier in result

Returns list of URIs matching the condition specified in fc

153.10.4.5 public abstract Promise<Boolean> notify(String uri, NotificationDTO notification)

uri uri of destination

notification content of notification

- send notification

Returns Promise of notification execution status

153.10.4.6 **public abstract Promise<ResponsePrimitiveDTO> request(RequestPrimitiveDTO request)**

request request primitive

- send a request and receive response.

This method allows very raw data type access and it enables all possible message exchanges among oneM2M entities. This is called the low level API. This method allows all possible operation of oneM2M. For the return type, OSGi Promise is used for allowing synchronous and asynchronous calling manner.

Returns promise of ResponseDTO.

153.10.4.7 **public abstract Promise<ResourceDTO> retrieve(String uri)**

uri URI for retrieving resource

- retrieve resource

retrieve resource on URI specified by uri argument. This method retrieve all attributes of the resource.

Returns retrieved resource data

153.10.4.8 **public abstract Promise<ResourceDTO> retrieve(String uri, List<String> targetAttributes)**

uri URI for retrieving resource

targetAttributes names of the target attribute

- retrieve resource with selected attributes.

retrieve resource on URI specified by uri argument. This method retrieve selected attributes by targetAttributes argument. The retrieve() methods are methods to retrieve resource on URI specified by uri argument.

Returns retrieved resource data

153.10.4.9 **public abstract Promise<ResourceDTO> update(String uri, ResourceDTO resource)**

uri URI for updating resource

resource data resource

- update resource

The update() method is a method to update resource on the URI specified by uri argument. The resource argument holds attributes to be updated. Attributes not to be updated shall not included in the argument.

Returns updated resource

153.11 **org.osgi.service.onem2m.dto**

Service Layer Data Transfer Objects for oneM2M Specification Package Version 1.0.

Bundles wishing to use this package must list the package in the Import-Package header of the bundle's manifest. This package has two types of users: the consumers that use the API in this package and the providers that implement the API in this package.

Example import for consumers using the API in this package:

```
{@code Import-Package: org.osgi.service.onem2m.dto; version="[1.0,2.0)"]}
```

Example import for providers implementing the API in this package:

```
{@code Import-Package: org.osgi.service.onem2m.dto; version="[1.0,1.1)"}
```

153.11.1

Summary

- AttributeDTO - DTO expresses Attribute.
- ChildResourceRefDTO - DTO expresses ChildResourceRef.
- Constants - This class defines constants for resource types.
- DasInfoDTO - DTO expresses DasInfo.
- FilterCriteriaDTO - DTO expresses FilterCriteria.
- GenericDTO - GenericDTO expresses miscellaneous data structures of oneM2M.
- IPEDiscoveryRequestDTO - IPEDiscoveryRequestDTO is an element of NotificationEventDTO
- LocalTokenIdAssignmentDTO - DTO expresses LocalTokenIdAssignment.
- NotificationDTO - DTO expresses Notification.
- NotificationEventDTO - DTO expresses NotificationEventDTO
- PrimitiveContentDTO - DTO expresses Primitive Content.
- ReleaseVersion - enum expresses oneM2M specification version.
- RequestPrimitiveDTO - DTO expresses Request Primitive.
- ResourceDTO - DTO expresses Resource.
- ResourceWrapperDTO - DTO expresses ResourceWrapper.
- ResponsePrimitiveDTO - DTO expresses Response Primitive.
- ResponseTypeInfoDTO - DTO expresses ResponseTypeInfo
- SecurityInfoDTO - DTO expresses Security Info.

153.11.2

public class AttributeDTO extends DTO

DTO expresses Attribute.

This class is typically used in FilterCriteriaDTO for expressing matching condition.

See Also oneM2M TS-0004 6.3.5.9 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.2.1

public String name

Attribute name

153.11.2.2

public Object value

Supposed value of the attribute

153.11.2.3

public AttributeDTO()

153.11.3

public class ChildResourceRefDTO extends DTO

DTO expresses ChildResourceRef.

See Also oneM2M TS-0004 6.3.5.29 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD childResourceRef [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L885-893]

Concurrency Not Thread-safe

153.11.3.1 public String name

name of the child resource pointed to by the URI

153.11.3.2 public String specializationID

resource type specialization of the child resource pointed to by the URI in case type represents a flexContainer. This is an optional field.

153.11.3.3 public Integer type

resourceType of the child resource pointed to by the URI

153.11.3.4 public String uri

URI to the child resource.

153.11.3.5 public ChildResourceRefDTO()

153.11.4 public final class Constants

This class defines constants for resource types.

See Also oneM2M TS-0004 6.3.4.2.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.4.1 public static final int RT_accessControlPolicy = 1

resource type for accessControlPolicy

153.11.4.2 public static final int RT_accessControlPolicyAnnc = 10001

resource type for accessControlPolicyAnnc

153.11.4.3 public static final int RT_AE = 2

resource type for AE

153.11.4.4 public static final int RT_AEAnnc = 10002

resource type for AEAnnc

153.11.4.5 public static final int RT_AEContactList = 43

resource type for AEContactList

153.11.4.6 public static final int RT_AEContactListPerCSE = 44

resource type for AEContactListPerCSE

153.11.4.7 public static final int RT_authorizationDecision = 35

resource type for authorizationDecision

153.11.4.8 public static final int RT_authorizationInformation = 37

resource type for authorizationInformation

153.11.4.9 public static final int RT_authorizationPolicy = 36

resource type for authorizationPolicy

153.11.4.10 public static final int RT_backgroundDataTransfer = 49

resource type for backgroundDataTransfer

153.11.4.11	public static final int RT_container = 3 resource type for container
153.11.4.12	public static final int RT_containerAnnc = 10003 resource type for containerAnnc
153.11.4.13	public static final int RT_contentInstance = 4 resource type for contentInstance
153.11.4.14	public static final int RT_contentInstanceAnnc = 10004 resource type for contentInstanceAnnc
153.11.4.15	public static final int RT_crossResourceSubscription = 48 resource type for crossResourceSubscription
153.11.4.16	public static final int RT_CSEBase = 5 resource type for CSEBase
153.11.4.17	public static final int RT_delivery = 6 resource type for delivery
153.11.4.18	public static final int RT_dynamicAuthorizationConsultation = 34 resource type for dynamicAuthorizationConsultation
153.11.4.19	public static final int RT_dynamicAuthorizationConsultationAnnc = 10034 resource type for dynamicAuthorizationConsultationAnnc
153.11.4.20	public static final int RT_eventConfig = 7 resource type for eventConfig
153.11.4.21	public static final int RT_execInstance = 8 resource type for execInstance
153.11.4.22	public static final int RT_flexContainer = 28 resource type for flexContainer
153.11.4.23	public static final int RT_flexContainerAnnc = 10028 resource type for flexContainerAnnc
153.11.4.24	public static final int RT_group = 9 resource type for group
153.11.4.25	public static final int RT_groupAnnc = 10009 resource type for groupAnnc
153.11.4.26	public static final int RT_localMulticastGroup = 45 resource type for localMulticastGroup
153.11.4.27	public static final int RT_locationPolicy = 10 resource type for locationPolicy

153.11.4.28	public static final int RT_locationPolicyAnnc = 10010 resource type for locationPolicyAnnc
153.11.4.29	public static final int RT_m2mServiceSubscriptionProfile = 11 resource type for m2mServiceSubscriptionProfile
153.11.4.30	public static final int RT_mgmtCmd = 12 resource type for mgmtCmd
153.11.4.31	public static final int RT_mgmtObj = 13 resource type for mgmtObj
153.11.4.32	public static final int RT_mgmtObjAnnc = 10013 resource type for mgmtObjAnnc
153.11.4.33	public static final int RT_multimediaSession = 46 resource type for multimediaSession
153.11.4.34	public static final int RT_multimediaSessionAnnc = 10046 resource type for multimediaSessionAnnc
153.11.4.35	public static final int RT_node = 14 resource type for node
153.11.4.36	public static final int RT_nodeAnnc = 10014 resource type for nodeAnnc
153.11.4.37	public static final int RT_notificationTargetMgmtPolicyRef = 25 resource type for notificationTargetMgmtPolicyRef
153.11.4.38	public static final int RT_notificationTargetPolicy = 26 resource type for notificationTargetPolicy
153.11.4.39	public static final int RT_ontology = 39 resource type for ontology
153.11.4.40	public static final int RT_ontologyAnnc = 10039 resource type for ontologyAnnc
153.11.4.41	public static final int RT_ontologyRepository = 38 resource type for ontologyRepository
153.11.4.42	public static final int RT_ontologyRepositoryAnnc = 10038 resource type for ontologyRepositoryAnnc
153.11.4.43	public static final int RT_policyDeletionRules = 27 resource type for policyDeletionRules
153.11.4.44	public static final int RT_pollingChannel = 15 resource type for pollingChannel

153.11.4.45	public static final int RT_remoteCSE = 16 resource type for remoteCSE
153.11.4.46	public static final int RT_remoteCSEAnnc = 10016 resource type for remoteCSEAnnc
153.11.4.47	public static final int RT_request = 17 resource type for request
153.11.4.48	public static final int RT_role = 31 resource type for role
153.11.4.49	public static final int RT_schedule = 18 resource type for schedule
153.11.4.50	public static final int RT_scheduleAnnc = 10018 resource type for scheduleAnnc
153.11.4.51	public static final int RT_semanticDescriptor = 24 resource type for semanticDescriptor
153.11.4.52	public static final int RT_semanticDescriptorAnnc = 10024 resource type for semanticDescriptorAnnc
153.11.4.53	public static final int RT_semanticMashupInstance = 41 resource type for semanticMashupInstance
153.11.4.54	public static final int RT_semanticMashupInstanceAnnc = 10041 resource type for semanticMashupInstanceAnnc
153.11.4.55	public static final int RT_semanticMashupJobProfile = 40 resource type for semanticMashupJobProfile
153.11.4.56	public static final int RT_semanticMashupJobProfileAnnc = 10040 resource type for semanticMashupJobProfileAnnc
153.11.4.57	public static final int RT_semanticMashupResult = 42 resource type for semanticMashupResult
153.11.4.58	public static final int RT_semanticMashupResultAnnc = 10042 resource type for semanticMashupResultAnnc
153.11.4.59	public static final int RT_serviceSubscribedAppRule = 19 resource type for serviceSubscribedAppRule
153.11.4.60	public static final int RT_serviceSubscribedNode = 20 resource type for serviceSubscribedNode
153.11.4.61	public static final int RT_statsCollect = 21 resource type for statsCollect

153.11.4.62 **public static final int RT_statsConfig = 22**

resource type for

153.11.4.63 **public static final int RT_subscription = 23**

resource type for statsConfig

153.11.4.64 **public static final int RT_timeSeries = 29**

resource type for timeSeries

153.11.4.65 **public static final int RT_timeSeriesAnnc = 10029**

resource type for timeSeriesAnnc

153.11.4.66 **public static final int RT_timeSeriesInstance = 30**

resource type for timeSeriesInstance

153.11.4.67 **public static final int RT_timeSeriesInstanceAnnc = 10030**

resource type for timeSeriesInstanceAnnc

153.11.4.68 **public static final int RT_token = 32**

resource type for token

153.11.4.69 **public static final int RT_transaction = 51**

resource type for transaction

153.11.4.70 **public static final int RT_transactionMgmt = 50**

resource type for transactionMgmt

153.11.4.71 **public static final int RT_triggerRequest = 47**

resource type for triggerRequest

153.11.5 **public class DasInfoDTO
extends DTO**

DTO expresses DasInfo. DAS is short for Dynamic Authorization Server.

See Also oneM2M TS-0004 6.3.5.45 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD dynAuthTokenReqInfo and dasInfo [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L1135-1147]

Concurrency Not Thread-safe

153.11.5.1 **public GenericDTO dasRequest**

Information to send to the Dynamic Authorization Server

See Also oneM2M XSD dynAuthDasRequest [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L1149-1198]

153.11.5.2 **public String securedDasRequest**

Secured Information to send to the Dynamic Authorization Server. JWS or JWE is assigned to this field.

153.11.5.3 **public String uri**

Dynamic Authorization Server URI

153.11.5.4 **public** DasInfoDTO()

153.11.6 **public class** FilterCriteriaDTO
extends DTO

DTO expresses FilterCriteria. This data structure is used for searching resources.

See Also oneM2M TS-0004 6.3.5.8 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oenM2M TS-0004 7.3.3.17.17 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.6.1 **public** String applyRelativePath

Apply Relative Path

153.11.6.2 **public** List<AttributeDTO> attribute

Attribute

153.11.6.3 **public** List<AttributeDTO> childAttribute

Child Attribute

153.11.6.4 **public** List<String> childLabels

Child Labels

153.11.6.5 **public** List<Integer> childResourceType

Child Resource Type

153.11.6.6 **public** String contentFilterQuery

Content Filter Query

153.11.6.7 **public** Integer contentFilterSyntax

Content Filter Syntax

153.11.6.8 **public** List<String> contentType

Content Type

153.11.6.9 **public** String createdAfter

Created After

153.11.6.10 **public** String createdBefore

Created Before

153.11.6.11 **public** String expireAfter

Expire After

153.11.6.12 **public** String expireBefore

Expire Before

153.11.6.13 **public** FilterOperation filterOperation

Filter Operation

153.11.6.14	public FilterUsage filterUsage Filter Usage
153.11.6.15	public List<String> labels Labels
153.11.6.16	public String labelsQuery Label Query
153.11.6.17	public Integer level Level
153.11.6.18	public Integer limit Limit number of Answers
153.11.6.19	public String modifiedSince Modified Since
153.11.6.20	public Integer offset Offset
153.11.6.21	public List<AttributeDTO> parentAttribute Parent Attribute
153.11.6.22	public List<String> parentLabels Parent Labels
153.11.6.23	public List<Integer> parentResourceType Parent Resource Type
153.11.6.24	public List<Integer> resourceType Resource Type
153.11.6.25	public List<String> semanticsFilter Semantic Filter
153.11.6.26	public Integer sizeAbove Size Above
153.11.6.27	public Integer sizeBelow Size Below
153.11.6.28	public Integer stateTagBigger State Tag Bigger
153.11.6.29	public Integer stateTagSmaller State Tag Smaller
153.11.6.30	public String unmodifiedSince Unmodified Since

153.11.6.31 **public FilterCriteriaDTO()**

153.11.7 **public class GenericDTO**
extends DTO

GenericDTO expresses miscellaneous data structures of oneM2M.

Concurrency Not Thread-safe

153.11.7.1 **public Map<String, Object> element**

Substructure of DTO. Type of the value part should be one of types allowed as OSGi DTO.

153.11.7.2 **public GenericDTO()**

153.11.8 **public class IPEDiscoveryRequestDTO**
extends DTO

IPEDiscoveryRequestDTO is an element of NotificationEventDTO

See Also oneM2M TS-0004 6.3.5.13 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.8.1 **public FilterCriteriaDTO filterCriteria**

FilterCriteria

See Also oneM2M TS-0004 6.3.5.8 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.8.2 **public String originator**

originator

153.11.8.3 **public IPEDiscoveryRequestDTO()**

153.11.9 **public class LocalTokenIdAssignmentDTO**
extends DTO

DTO expresses LocalTokenIdAssignment.

See Also oneM2M XSD dynAuthLocalTokenIdAssignments and localTokenIdAssignment [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L1112-1123]

Concurrency Not Thread-safe

153.11.9.1 **public String localTokenID**

local token ID

153.11.9.2 **public String tokenID**

token ID

153.11.9.3 **public LocalTokenIdAssignmentDTO()**

153.11.10 **public class NotificationDTO**
extends DTO

DTO expresses Notification.

See Also oneM2M TS-0004 6.3.5.13 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.10.1 public Boolean aeReferenceIDChange

aeReferenceIDChange element

See Also oneM2M TS-0004 7.5.1.2.17 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.10.2 public Boolean aeRegistrationPointChange

AE Registration Point Change

See Also oneM2M TS-0004 7.5.1.2.16 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.10.3 public String creator

creator

153.11.10.4 public IPEDiscoveryRequestDTO ipeDiscoveryRequest

IPE Discovery Request.

153.11.10.5 public NotificationEventDTO notificationEvent

Notification Event

153.11.10.6 public String notificationForwardingURI

notification forwarding URI

153.11.10.7 public String notificationTarget

ID for notification target

153.11.10.8 public Boolean subscriptionDeletion

Flag showing subscription deletion This field is optional.

153.11.10.9 public String subscriptionReference

URI referring subscription resource.

153.11.10.10 public String trackingID1

tracking ID 1

See Also oneM2M TS-0004 7.5.1.2.16 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.5.1.2.17 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.10.11 public String trackingID2

tracking ID 1

See Also oneM2M TS-0004 7.5.1.2.16 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.5.1.2.17 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.10.12 public Boolean verificationRequest

Flag showing verification request. This field is optional.

153.11.10.13 **public NotificationDTO()**

153.11.11 **public class NotificationEventDTO**

DTO expresses NotificationEventDTO

This data structure is held in NotificationDTO.

See Also oneM2M TS-0004 6.3.5.13 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.11.1 **public NotificationEventType notificationEventType**

notificationEventType

See Also oneM2M TS-0004 6.3.4.2.19 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.11.2 **public Map<String, Object> operationMonitor**

operationMonitor

See Also oneM2M TS-0004 6.3.5.57 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.11.3 **public Object representation**

m2m:representation

See Also oneM2M TS-0004 6.3.5.62 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.11.4 **public NotificationEventDTO()**

153.11.12 **public class PrimitiveContentDTO**
extends DTO

DTO expresses Primitive Content.

This Data structure is used as union. Only one field MUST have a value, the others MUST be null.

See Also oneM2M TS-0004 6.3.5.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.2.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD primitiveContent [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-common-Types-v3_11_0.xsd#L596-602]

Concurrency Not Thread-safe

153.11.12.1 **public List<NotificationDTO> aggregatedNotification**

Aggregated Notification

153.11.12.2 **public List<ResponsePrimitiveDTO> aggregatedResponse**

Aggregated Response

153.11.12.3 **public List<String> attributelist**

Attribute List

153.11.12.4 **public List<ChildResourceRefDTO> childResourceRefList**

Child Resource RefList

- 153.11.12.5** **public String debugInfo**
Debug Info
- 153.11.12.6** **public List<String> listOfURIs**
List Of URIs
- 153.11.12.7** **public NotificationDTO notification**
Notification
- 153.11.12.8** **public String queryResult**
Query Result
- 153.11.12.9** **public RequestPrimitiveDTO requestPrimitive**
Request Primitive
- 153.11.12.10** **public ResourceDTO resource**
Resource
- 153.11.12.11** **public ResourceWrapperDTO resourceWrapper**
Resource Wrapper
- 153.11.12.12** **public ResponsePrimitiveDTO responsePrimitive**
Response Primitive
- 153.11.12.13** **public SecurityInfoDTO securityInfo**
Security Info
- 153.11.12.14** **public String uri**
URI
- 153.11.12.15** **public PrimitiveContentDTO()**

153.11.13 **enum ReleaseVersion**

enum expresses oneM2M specification version.

This information is introduced after Release 2.0 and oneM2M uses only R2A, R3_0 (as 2a and 3).

See Also oneM2M XSD releaseVersion [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L450-455]

- 153.11.13.1** **R1_0**
Release 1
- 153.11.13.2** **R1_1**
Release 1.1
- 153.11.13.3** **R2_0**
Release 2
- 153.11.13.4** **R2A**
Release 2A

153.11.13.5 R3_o

Release 3

153.11.13.6 R4_o

Release 4 (reserved for future)

153.11.13.7 R5_o

Release 5 (reserved for future)

153.11.13.8 public static ReleaseVersion valueOf(String name)**153.11.13.9 public static ReleaseVersion[][] values()****153.11.14 public class RequestPrimitiveDTO
extends DTO**

DTO expresses Request Primitive.

See Also oneM2M TS-0004 6.4.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD requestPrimitive [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-requestPrimitive-v3_11_0.xsd]

Concurrency Not Thread-safe

153.11.14.1 public Boolean authorizationRelationshipIndicator

Authorization Relationship Indicator

153.11.14.2 public Boolean authorizationSignatureIndicator

Authorization Signature Indicator

153.11.14.3 public List<String> authorizationSignatures

Authorization Signatures

In oneM2M this parameter is expressed in m2m:signatureList.

See Also oneM2M TS-0004 6.3.4.2.8 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD signatureList [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L129-137]

153.11.14.4 public PrimitiveContentDTO content

Primitive Content

See Also oneM2M TS-0004 6.3.5.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.2.1.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.5 public Boolean deliveryAggregation

Delivery Aggregation

This parameter is related to CMDH(Communication Management and Delivery Handling) policy.

See Also oneM2M TS-0004 D.12 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.6 public DesiredIdentifierResultType desiredIdentifierResultType

Desired Identifier Result Type

This parameter specifies identifier type in response, such as structured or unstructured. This parameter used to be Discovery Result Type in previous oneM2M release.

See Also oneM2M TS-0004 6.3.4.2.8 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.7 public Integer eventCategory

Event Category

allowed values are 2(Immediate), 3(BestEffort), 4(Latest), and 100-999 as user defined range.

See Also oneM2M TS-0004 6.3.3 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD eventCat [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L326], oneM2M XSD stdEventCats [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-enumerationTypes-v3_11_0.xsd#L208-221]

153.11.14.8 public FilterCriteriaDTO filterCriteria

Filter Criteria

See Also oneM2M TS-0004 6.3.5.8 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.9 public String from

From Parameter.

Originator of the request is stored.

See Also oneM2M TS-0004 6.3.4.2.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.10 public String groupRequestIdentifier

Group Request Identifier TODO: search doc.

153.11.14.11 public List<String> groupRequestTargetMembers

Group Request Target Members

153.11.14.12 public List<String> localTokenIDs

Local Token Identifiers

In oneM2M this parameter is expressed as list of xs:NCName.

153.11.14.13 public Operation operation

Operation This field is mandatory.

See Also oneM2M TS-0004 6.3.4.2.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.14 public String operationExecutionTime

Operation Execution Time

153.11.14.15 public String originatingTimestamp

Originating Timestamp

153.11.14.16 public ReleaseVersion releaseVersionIndicator

Release Version

153.11.14.17 public String requestExpirationTimestamp

Request Expiration Timestamp

* This parameter is related to CMDH(Communication Management and Delivery Handling) policy.

See Also oneM2M TS-0004 D.12 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.18 public String requestIdentifier

Request Identifier

See Also oneM2M TS-0004 6.3.3 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.19 public Integer resourceType

Resource Type

See Also oneM2M TS-0004 6.3.4.2.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.20 public ResponseTypeInfoDTO responseType

Response Type Info

See Also oneM2M TS-0004 6.3.5.30 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.21 public ResultContent resultContent

Result Content

See Also oneM2M TS-0004 6.3.4.2.7 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.22 public String resultExpirationTimestamp

Result Expiration Timestamp

This parameter is related to CMDH(Communication Management and Delivery Handling) policy.

See Also oneM2M TS-0004 D.12 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.23 public String resultPersistence

Result Persistence

This parameter is related to CMDH(Communication Management and Delivery Handling) policy.

See Also oneM2M TS-0004 D.12 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.24 public List<String> roleIDs

Role IDs

153.11.14.25 public Boolean semanticQueryIndicator

Semantic Query Indicator

See Also oneM2M TS-0004 7.3.3.19 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.14.26 public String to

To Parameter

153.11.14.27 public List<String> tokenIDs

Token Identifiers

In oneM2M this parameter is expressed as list of m2m:tokenID.

See Also oneM2M XSD signatureList [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L66-70]

153.11.14.28 public Boolean tokenRequestIndicator

Token Request Indicator

153.11.14.29 public List<String> tokens

Tokens

Each token is in m2m:dynAuthJWT

See Also oneM2M XSD signatureList [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L417-426]

153.11.14.30 public String vendorInformation

Vendor Information

Used for vendor specific information. No procedure is defined for the parameter.

153.11.14.31 public RequestPrimitiveDTO()**153.11.15 public class ResourceDTO
extends DTO**

DTO expresses Resource.

Universal attributes are expressed in field of the class. Common attributes and other attributes are stored in attribute field.

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

Concurrency Not Thread-safe

153.11.15.1 public Map<String, Object> attribute

Non Universal Attribute. Value Part must be the types that are allowed for OSGi DTO. In case of value part can be expressed DTO in this package, the DTO must be used. In case of value part have sub-elements, GenericDTO must be used.

153.11.15.2 public String creationTime

Creation time

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

153.11.15.3 public String lastModifiedTime

last modified time

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

153.11.15.4 public String parentID

Parent ID Resource ID of parent resource.

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

153.11.15.5 public String resourceID

Resource ID

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

153.11.15.6 public String resourceName

Resource name

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf]

153.11.15.7 public Integer resourceType

Resource Type

See Also oneM2M TS-0001 9.6.1.3.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf], oneM2M TS-0004 6.3.4.2.1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.15.8 public ResourceDTO()

153.11.16 public class ResourceWrapperDTO extends DTO

DTO expresses ResourceWrapper.

See Also oneM2M TS-0004 6.3.5.25 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.16.1 public ResourceDTO resource

Resource

153.11.16.2 public String uri

Hierarchical URI of the resource

153.11.16.3 public ResourceWrapperDTO()

153.11.17 public class ResponsePrimitiveDTO extends DTO

DTO expresses Response Primitive.

See Also oneM2M TS-0004 6.4.2 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD responsePrimitive [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-responsePrimitive-v3_11_0.xsd]

Concurrency Not Thread-safe

153.11.17.1 public List<LocalTokenIdAssignmentDTO> assignedTokenIdIdentifiers

Assigned Token Identifiers

See Also oneM2M TS-0004 6.3.5.43 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

- 153.11.17.2** **public Boolean AuthSignatureReqInfo**
AuthSignatureReqInfo
- 153.11.17.3** **public PrimitiveContentDTO content**
Primitive Content
See Also oneM2M TS-0004 6.3.5.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.2.1.2 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.17.4** **public Integer contentOffset**
Content Offset
- 153.11.17.5** **public ContentStatus contentStatus**
Content Status
See Also oneM2M TS-0004 6.3.4.2.44 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.17.6** **public Integer eventCategory**
Event Category
allowed values are 2(Immediate), 3(BestEffort), 4(Latest), and 100-999 as user defined range.
See Also oneM2M TS-0004 6.3.3 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M XSD eventCat [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-commonTypes-v3_11_0.xsd#L326], oneM2M XSD stdEventCats [https://git.onem2m.org/PRO/XSD/blob/master/v3_11_0/CDT-enumerationTypes-v3_11_0.xsd#L208-221]
- 153.11.17.7** **public String from**
From Parameter
- 153.11.17.8** **public String originatingTimestamp**
Originating Timestamp To Parameter
See Also oneM2M TS-0004 Table 6.3.3-1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.17.9** **public ReleaseVersion releaseVersionIndicator**
Release Version Indicator
- 153.11.17.10** **public String requestIdentifier**
Request Identifier
See Also oneM2M TS-0004 6.3.3 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.17.11** **public Integer responseStatusCode**
Response Status Code
See Also oneM2M TS-0004 6.3.4.2.9 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.17.12** **public String resultExpirationTimestamp**
ResultExpiration Timestamp

See Also oneM2M TS-0004 Table 6.3.3-1 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.17.13 **public String to**

To Parameter

See Also oneM2M TS-0004 6.3.3 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.17.14 **public List<DasInfoDTO> tokenReqInfo**

Token Request Info

See Also oneM2M TS-0004 6.3.5.45 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.17.15 **public String vendorInformation**

Vendor Information

Used for vendor specific information. No procedure is defined for the parameter.

153.11.17.16 **public ResponsePrimitiveDTO()**

153.11.18 **public class ResponseTypeInfoDTO extends DTO**

DTO expresses ResponseTypeInfo

See Also oneM2M TS-0004 6.3.5.30 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

153.11.18.1 **public List<String> notificationURI**

Notification URI

See Also oneM2M TS-0004 6.3.5.30 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf], oneM2M TS-0004 7.5.1.2.5 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.18.2 **public ResponseType responseTypeValue**

Response Type Value

See Also oneM2M TS-0004 6.3.4.2.6 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

153.11.18.3 **public ResponseTypeInfoDTO()**

153.11.19 **public class SecurityInfoDTO extends DTO**

DTO expresses Security Info.

This class is used as union. SecurityInfoType field indicates which type of content is stored.

See Also oneM2M TS-0004 6.3.5.48 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]

Concurrency Not Thread-safe

- 153.11.19.1** **public GenericDTO dasRequest**
Das Request
- 153.11.19.2** **public GenericDTO dasResponse**
Das Response
- 153.11.19.3** **public byte[][] escertkeMessage**
Escertke Message
- 153.11.19.4** **public String esprimObject**
Esprim Object
- 153.11.19.5** **public GenericDTO esprimRandObject**
Esprim Rand Object
- 153.11.19.6** **public SecurityInfoType securityInfoType**
Security Info Type
- See Also* oenM2M TS-0004 6.3.4.2.35 [http://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf]
- 153.11.19.7** **public SecurityInfoDTO()**

153.12 References

- [1] *oneM2M: TS-0001 Functional Architecture V3.15.1*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0001-Functional_Architecture-V3_15_1.pdf
- [2] *oneM2M: TS-0004 Service Layer Core Protocol V3.11.2*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0004_Service_Layer_Core_Protocol_V3_11_2.pdf
- [3] *oneM2M: TS-0008 CoAP Protocol Binding V3.3.1*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0008-CoAP_Protocol_Binding-V3_3_1cl.pdf
- [4] *oneM2M: TS-0009 HTTP Protocol Binding V3.2.1*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0009-HTTP_Protocol_Binding-V3_2_1cl.pdf
- [5] *oneM2M: TS-0010 MQTT Protocol Binding V3.0.1*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0010-MQTT_protocol_binding-V3_0_1.pdf
- [6] *oneM2M: TS-0020 WebSocket Protocol Binding V3.0.1*
https://www.onem2m.org/images/files/deliverables/Release3/TS-0020-WebSocket_Protocol_Binding-V3_0_1.pdf
- [7] *oneM2M-schemas*
<https://git.onem2m.org/PRO/XSD>

DRAFT

Licensed under the Eclipse Foundation Specification License – v1.0. Copyright © Contributors to the Eclipse Foundation

End Of Document

Licensed under the Eclipse Foundation Specification License – v1.0. Copyright © Contributors to the Eclipse Foundation

DRAFT